Advances in Computational Collective Intelligence

COMPUTATIONAL INTELLIGENCE IN INDUSTRY 4.0 AND 5.0 APPLICATIONS

Trends, Challenges and Applications

Edited by
Joseph Bamidele Awotunde,
Kamalakanta Muduli and Biswajit Brahma

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Computational Intelligence in Industry 4.0 and 5.0 Applications

Industry 4.0 and 5.0 applications will revolutionize production, enabling smart manufacturing machines to interact with their environments. These machines will become self-aware, self-learning, and capable of real-time data interpretation for self-diagnosis and prevention of production issues. They will also self-calibrate and prioritize tasks to enhance production quality and efficiency.

Computational Intelligence in Industry 4.0 and 5.0 Applications examines applications that merge three key disciplines: computational intelligence (CI), Industry 4.0, and Industry 5.0. It presents solutions using Industrial Internet of Things (IIoT) technologies, augmented by CI-based techniques, modeling, controls, estimations, applications, systems, and future scopes. These applications use data from smart sensors, processed through enhanced CI methods, to make smart automation more effective.

Industry 4.0 integrates data and intelligent automation into manufacturing, using technologies like CI, the IoT, the IIoT, and cloud computing. It transforms data into actionable insights for decision-making and process optimization, essential for modern competitive businesses managing high-speed data integration in production processes. Currently, Industries 4.0 and 5.0 are undergoing significant transformations due to advances in applying artificial intelligence (AI), big data analytics, telecommunication technologies, and control theory. These applications are increasingly multidisciplinary, integrating mechanical, control, and information technologies. However, they face such technical challenges as parametric uncertainties, external disturbances, sensor noise, and mechanical failures. To address these, this book examines such CI technologies as fuzzy logic, neural networks, and reinforcement learning and their application to modeling, control, and estimation. It also covers recent advancements in IIoT sensors, microcontrollers, and big data analytics that further enhance CI-based solutions in Industry 4.0 and 5.0 systems.

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1 Industry 4.0

Challenges and Issues Encountered in Manufacturing and Testing of MEMS Devices

Hemant Narayan, M. Prakash, B. Shakila, R. Kumar, and Sivakumar Annamalai Palanivel

1.1 INTRODUCTION

Industry 4.0 has been discussed since the beginning of the century. The term "Industry 4.0" refers to the continual growth of the manufacturing and industrial sectors via the incorporation of digital technology, automation, and data exchange. It builds on the preceding three industrial revolutions, which resulted in substan tial advances in manufacturing processes. There are still some unclear areas concerning the implications and repercussions of this new manufacturing approach.

Industry 4.0 is characterized by the convergence of various technologies, including the IoT (Internet of Things), big data analytics, robotics, AI (artificial intelligence), and cloud computing. These technologies enable the creation of "smart factories" and facilitate the digitalization and interconnectedness of the entire value chain. It is built on the use of cutting-edge monitoring, information management, and transmission systems to enhance ongoing manufacturing processes [1]. Yet, despite increasing curiosity in the Industry 4.0 area, it still represents a quasi-idea. The adoption of Industry 4.0 technologies offers several potential benefits, including increased productivity, improved efficiency, enhanced flexibility, better quality control, reduced costs, and the ability to respond quickly to market demands. Industry 4.0 offers opportunities for innovation, improved efficiency, and enhanced functionality in the development, manufacturing, and utilization of MEMS devices. It enables the integration of MEMS into the digital ecosystem, fostering the growth of smart, interconnected systems and applications. Although it is the case that MEMS technology has evolved from integrated circuits fabrication techniques, these technologies' test approaches

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are distinct. Because of the immersion of several domains, MEMS devices are complicated and diversified.

MEMS has exceptional adaptability and malleability in the miniaturization of devices, with the prospects of utilizing less energy and operating as envisioned. Designing smaller, lighter, and more powerful devices is now a popular micro/nano system trend [2].

1.2 KEY CONTRIBUTIONS OF THIS CHAPTER

The following are the most significant contributions of this chapter.

1. The chapter provides immense knowledge about Industry 4.0 and MEMS including its constituents, history, and applications.
2. The significant issues and challenges faced during MEMS manufacturing, testing, and packaging, in general, are discussed.
3. Biomedical applications of MEMS devices and their challenges are discussed in depth.

1.3 CHAPTER ORGANIZATION

Section 1.4 discusses a literature review for industry revolution and key components of Industry 4.0. Section 1.5 elaborates MEMS and its constituents, while Section 1.6 discusses the principles of the piezoelectric effect. The history of piezoelectric materials is discussed in Section 1.7 and Section 1.8 illustrates the key components of Industry 4.0 in MEMS. Section 1.9 depicts various methodologies of testing MEMS along with their breakdown and causes. Section 1.10 describes the difficulties encountered during the fabrication of MEMS devices and Section 1.11 describes MEMS testing concerns such as molding, dicing, designing, and wafer level issues. Section 1.12 depicts biomedical applications of MEMS devices and their issues and challenges. Lastly, Section 1.13 provides findings and gives research proposals for the future.

1.4 LITERATURE REVIEW

1.4.1 Industrial Revolution

Industry 3.0 represents the era of computerization and the use of electronics and IT in industrial processes. Key features include the introduction of computers, automation of production processes, and the use of early robotics. Industry 4.0, also known as the 4th Industrial Revolution, is characterized by the integration of smart technology, the Internet of Things (IoT), and cyber-physical systems into the manufacturing environment. It involves the extensive use of data, connectivity, and automation. Industry 5.0 is a concept that looks beyond pure automation and emphasizes the collaboration between humans and machines. It focuses on human-centric approaches, with AI and automation supporting human workers rather than replacing them entirely.

1.4.2 Key Components of Industry 4.0

Cyber-Physical Systems (CPS)

Machines and sensors are linked to the digital world, allowing for real-time monitoring, surveillance, and optimization of production processes.

Internet of Things (IoT)

Data are collected and exchanged by connected devices and sensors, allowing machines, goods, and systems to communicate with one another. The Internet of Things allows machines and things to interact, share information, and carry out activities without the need for direct human intervention. This collection of networked devices extends traditional computing's capabilities by allowing physical items to perceive, evaluate, and respond to their surroundings, making them "smart" and capable of interacting with the digital world.

Big Data Analytics

Large amounts of primary data acquired from many sources are analyzed in order to extract valuable information, optimize processes, and make educated choices.

Cloud Computing

Cloud-based services and platforms provide storage, computing power, and collaboration tools to support data processing and enable remote access to resources. Cloud computing is a model of computing that involves the delivery of computing resources and services over the internet ("the cloud"). Instead of hosting applications, data, and services on local servers or personal computers, cloud computing enables users to access and use these resources remotely via the internet. Cloud computing has revolutionized the way businesses and individuals utilize and manage IT resources, enabling more efficient, scalable, and cost-effective computing solutions.

ML (Machine Learning) and AI (Artificial Intelligence)

Machine learning is a branch of artificial intelligence (AI) that concentrates on the development of algorithms and statistical models that allow computers to gather information from data and make forecasts or assessments without being specifically programmed. Machine learning's major purpose is to enable computers to enhance their efficacy on a given activity through experience, despite human intervention. In machine learning, algorithms are trained on a dataset, which consists of input data (features) and corresponding output labels (targets). During the training process, the algorithm learns patterns and relationships in the data to make predictions or judgments on novel, unseen data.

AI is the imitation of human intellect in computers that are programmed to do tasks that would typically need human cognitive abilities. AI aims to create computer systems that can mimic human thinking, reasoning, learning, and problem-solving processes to perform tasks more efficiently and accurately. Machine learning, robotics, expert systems, computer vision, natural language processing, and other techniques and approaches are all part of the subject of artificial intelligence. These technologies allow AI systems to gather and analyze enormous volumes of data, detect patterns, forecast outcomes, and change their behavior depending on past experience.

Additive Manufacturing (3D Printing)

The ability to create complex objects layer by layer using 3D printing technology enables fast prototyping, customization, and decentralized production.

Virtual Reality (VR)

VR refers to a computer-generated simulation or a virtual environment that immerses users in a fully artificial, computer-generated 3D world. VR devices, such as head-mounted displays (HMDs) or VR goggles, block out the physical world and replace it with a digitally created environment, making users feel like they are present and interacting within that simulated world. Users can explore and interact with the virtual environment through head and hand movements, enabling a sense of presence and interaction with virtual objects and spaces. VR is commonly used in gaming, training, simulations, education, architecture, and various other fields where a realistic and immersive experience is desired.

Augmented Reality (AR)

Augmented reality overlays digital information onto the physical-world environment, enhancing or augmenting the user's perception of reality. AR technology allows users to see and interact with digital elements while still being aware of and able to interact with the real world. AR is typically experienced through smartphones, tablets, smart glasses, or other wearable devices that use cameras to capture the real-world scene and superimpose computer-generated graphics, images, or information on top of it. AR gives us a wide range of applications, including mobile gaming, navigation, industrial training, remote assistance, marketing, and interactive user manuals.

1.5 MICROELECTROMECHANICAL SYSTEMS

Microelectromechanical systems (MEMS) have undergone a rapid ascent to prominence as a vital technology over the last 10 years. MEMS is based on the greater volume efficiency and lower unit prices that the microelectronics industry has achieved in the past 60 years which may be transformed into devices that integrate mechanical and electrical components into a single silicon chip. Figure 1.1 illustrates the constituents of MEMS.

The majority of Si-based components, conductive metals, and some organic components have been used in MEMS devices since their functionality was mostly derived from intricate microstructures created by Si microfabrication processes. Functional materials, contrary to common belief, such as ferroelectric materials, are a form of piezoelectric material with impulsive electric polarization that may be inverted by an applied electric field. These materials are characterized by their ability to maintain a stable polarization state even after the external electric field is removed, making them useful for a broad area of applications, including sensors, actuators, memory devices, and energy-harvesting devices.

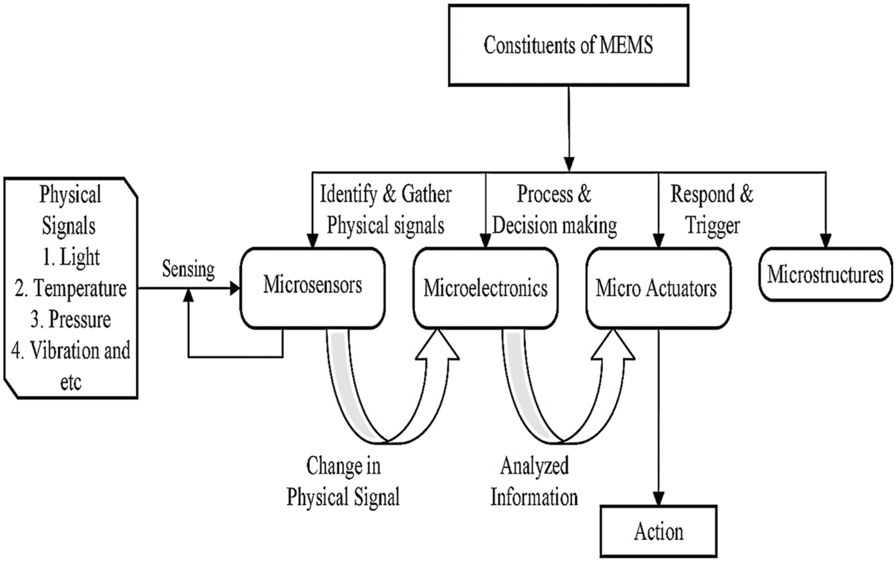

FIGURE 1.1 Constituents of MEMS [3].

These materials are typically crystalline in nature, and their unique properties arise from the asymmetry in their crystal structure, which leads to the presence of permanent electric dipoles within the material. The electric dipoles can align in multiple stable directions, resulting in the material having multiple polarization states or domains. These materials exhibit hysteresis behavior, which means that the polarization-versus-electric-field curve forms a loop when plotted. This hysteresis loop represents the ability of the material to switch between different polarization states with the application of an electric field and retain the polarization state after the field is removed. This property makes ferroelectric materials suitable for applications where stable and reversible polarization states are required.

Some common examples of ferroelectric materials include lithium niobate ($LiNbO_3$), lead zirconate titanate (PZT), barium titanate ($BaTiO_3$), and potassium niobate ($KNbO_3$), among others. These materials are widely used in various technologies, including sensors for pressure, acceleration, and temperature, ultrasonic transducers, actuators for precise motion control, nonvolatile ferroelectric memories, and energy-harvesting devices for converting mechanical vibrations or thermal gradients into electrical energy. Among these, piezoelectricity is very attractive in the application of microsensors and microactuators.

The implementation of digital technologies that can collect data in real time and analyze it is essential to the success of Industry 4.0. Providing the production system with information that is helpful to it in real time, temperature sensors, displacement sensors, pressure sensors, vibration sensors, torque sensors, power flow rate sensors, and variable sensors all make measurements that enable the creation of a digital representation of the manufacturing process. The objectives of the application should guide both the selection of sensors and the architecture of signal processing. In contemporary technology, MEMS sensors—such as the PulseNG sensor—significantly contribute to the measurement of acceleration. These sensors play a crucial role in monitoring signals. The ability to be manufactured in micro sizes, the requirement for less power, and also possibility of their integrated circuits and their usefulness, enable innovation in consumer electronics such as smartphones, gaming controllers, and similar devices. Accelerometers and gyroscopes made with MEMS technology are finding applications in a wide variety of industrial systems. MEMS sensors, designed for diagnostic purposes, should be permanently integrated into the device.After the sensor has been installed, a cable is used to link it to the IBU–NG interface, which is the housing for the electronic board.

This connection is later used to connect the sensor to the PLC controller of the machine, allowing both components to operate at the same time. Electrical components are responsible for reading the pulses that are produced by the MEMS sensor. The interface bus unit will be scheduled based on the data collected from the MEMS sensor, which monitors restrictions and vibrations. Once this information is gathered, the necessary settings will be configured programmatically.The policies governing maintenance have a significant impact on the rate of quality improvement. The manufacturers of vibration-monitoring and total productive maintenance/total quality management (TPM/TQM) systems will contribute to an increase in the life cycle of machine tools, which are employed in with high-quality performance. The MEMS sensor detects and quantifies vibrations by measuring acceleration on both the machine table and the spindle housing. These data are then used for information and predictive analysis [1]. Recently, countless research works have involving the applications of MEMS in the biomedical field, especially in microsurgery, microtherapy, artificial body parts, drug synthesis and delivery, and cell manipulation and characterization.

Piezoelectric energy harvesting is a sophisticated technology with great potential, and the performance of energy harvesting can be improved by adjusting a few important group parameters in terms of ceramic, polymer, single-crystal composite materials, and lead-free materials have their benefits and drawbacks. When it comes to piezoelectric polymers, PVDF-TrFE (polyvinylidene fluoride—trifluoroethylene) stands out as the most commonly utilized material among both PVDF and copolymers. Common PZT ceramics are made with hazardous lead. Consequently, it is important to look at potential alternatives to meet environmental regulations for new uses of existing materials. Consideration must also be given to the stability and state of the environment. Binding vibration mode is simple to tune at low frequency, and is used in these energy harvesters. High-voltage generators can benefit more from the impact design. Multiple potential ignition mechanisms have been investigated. Energy harvesters could one day power everything from personal electronics to entire smart cities, not to mention a wide variety of other uses in industries and the military. To fully utilize the potential of piezoelectric as sustainable energy sources, it will be necessary to optimize materials, make discoveries, create low-power devices, explore energy sources, increase energy output, and match applications [4].

Integrated circuits were employed in the development of devices and systems based on microelectromechanical systems (MEMS). These systems encompass design, fabrication techniques, and applications, including sensors and actuators. Procedures for processing in batches that are compatible, and sizes that span from micrometers to millimeters, are being created. MEMS technology derives from the tools and processes used to fabricate integrated circuits. Thermal sensors, pressure sensors, resonators, levers, and gear and gearbox systems, all of which are measured in microns, have the potential to be batch built together on the same chip as the processing unit. They make up a system on a chip in its own right, having numerous applications in the commercial sector. There is microfabrication of huge numbers of devices, each of which is capable of performing a single straightforward function. However, when combined, they are capable of performing complex functions. Not only do MEMS involve a reduction in the size of mechanical systems, they also constitute a novel approach to the design of mechanical devices and structures. Integrated microelectronic circuits are sometimes referred to as the "brains" of a device. There are several different types of MEMS devices, including sensors, actuators, bio-MEMS, microfluidic MEMS, RF MEMS, and optical MEMS. This forthcoming technological revolution of miniaturization known as MEMS has been acknowledged as providing valuable goods for a variety of commercial and government applications. The construction of micromachines that have small dimensions, low power consumption, and exquisite performance can only be accomplished with the assistance of devices that use MEMS technology [5].

1.6 PRINCIPLE OF THE PIEZOELECTRIC EFFECT

There are two distinct characteristics of piezoelectricity, direct and indirect piezoelectric effects (PE). The first effect is crucial for sensing and energy harvesting when a material is polarized under applied stress to procreate surface charges on the piezoelectric materials. In the second effect, electric potential induces mechanical strain or mechanical displacement in the material [6].

The piezoelectric constitutive equation shown below governs the direct and inverse piezoelectric effects.

$$\begin{bmatrix} S \\ D \end{bmatrix} = \begin{bmatrix} \alpha^E & \beta^t \\ \beta & \gamma^T \end{bmatrix} \begin{bmatrix} \mathrm{T} \\ \mathrm{E} \end{bmatrix} \tag{1.1}$$

This illustrates the relationship between strain (S), stress (T), electric displacement (D), and electric field (E) in the context of piezoelectric materials. It introduces parameters such as elastic compliance (α), piezoelectric coefficient (β), and dielectric constant (γ). Notably, superscripts E and T signify that certain constants are calculated at unaltered electric field and stress, respectively, while the superscript "t" indicates the transpose [7,8]. It highlights the impact of the direction of applied stress on the energy-harvesting performance of piezoelectric elements, emphasizing that for many piezoelectric materials, the polar axis is clearly defined. The direction of applied stress relative to this polar axis becomes crucial in determining the efficiency of energy scavenging. The text further notes that in ferroelectric ceramics or polymers like PVDF or PZT, the polar axis depends on the poling direction. On the other hand,

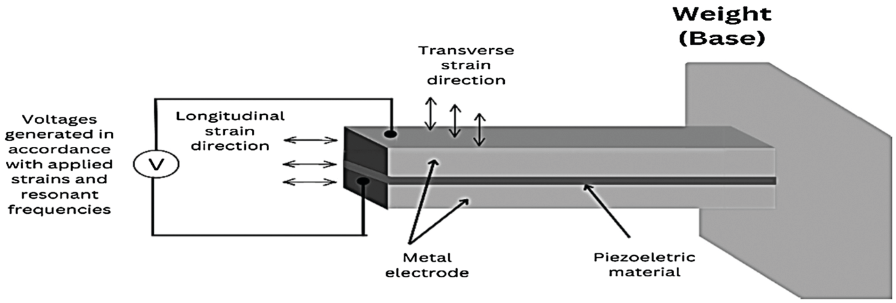

FIGURE 1.2 Mechanical work flows of piezoelectricity [8].

for non-ferroelectric crystalline materials such as ZnO or aluminum nitride (AlN), the polar axis is determined by the crystal orientation, specifically along the c-axis of the wurtzite crystal structure [9].

The mechanical function of piezoelectricity is shown in Figure 1.2, in which the transverse strain direction and longitudinal strain direction are mentioned and also the direction in which voltage is generated in accordance with applied strains and resonant frequencies.

1.7 PIEZOELECTRIC MATERIALS: HISTORY

The performance of micropiezoelectric materials over the last 10 years is shown in Figure 1.3. An assessment of various enhanced techniques is presented for cantilevers such as maximum resonant frequency cantilever, the 2-D cantilever beam with weight, the optimal device volume, and the four doubly clamped beams, including silicon-on-insulator (SOI) structures such as the addition of a SiO_2 layer and the bonding of bulk PZT oxide. It appears that significant searches are also being conducted for well-known piezoelectric materials such as potassium/sodium/niobium oxide (KNN), sodium niobium oxide ($NaNbO_3$) nanowires, AlN composites, PZT nanowires, $BaTiO_3$ nanogenerators, ZnO nanowire nanogenerators, and KNN/manganese oxide (Mn)/potassium (K) copper (C) tantalum (T) oxide (KCT) materials. Figure 1.3 compares various distinct methods of improvement, such as the hybrid model of electromagnetic-piezoelectric modifications like curved L-shaped and Si seismic weights on a uni-morph cantilever, and ad hoc options such as a blender caping with axial force and 12 bimorph actuators on a wind turbine [10].

As illustrated in Figure 1.3, the progressive advancement of micropiezoelectric energy harvesters over past 10 years demonstrates a favorable reaction to the system's potential. Power densities were obtained by using maximum resonant frequency cantilever, a two-dimensional cantilever beam with seismic mass, and optimum device volumes of 0.16, 0.24, and 0.31 mW/cm^3 were attained in the experiments. These findings were attained as a result of the cantilever constructions' low resonance frequencies, which were further minimized by the presence of mass at the cantilever's end. These piezoelectric devices frequently focus on low-frequency motions of the human body for their applications. In order to improve the resonance frequencies over a larger operational bandwidth and to

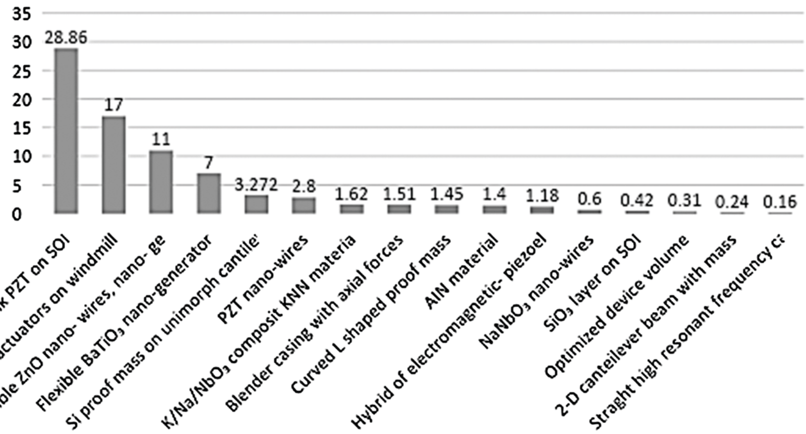

FIGURE 1.3 Power density (mWcm^3) of piezoelectricity systems in the last decade [8].

offer higher output power, more precise piezoelectric structures are needed. This would make them appropriate for high-frequency applications. Lower Curie temperatures, lower relative densities, and decreased poling activity in perovskite-type ferroelectric $BaTiO_3$ materials all have the potential to diminish the piezoelectric performance. Its relatively high piezoelectric constant, however, may allow for its inclusion in piezoelectric modules. Using MEMS and soft lithography, Hwang [10] effectively created a flexible $BaTiO_3$ nanogenerator with up to 7 mW/cm^3 of power conversion. Empirical studies on a flexible nanogenerator based on ZnO nanowires that was made using a scalable sweeping-printing technique were conducted. The power harvesting of the nanogenerator in this case scaled up to 11 mW/cm^3, a value that is 12–22 times higher than that obtained by the standard PZT cantilever system [11,12].

In an experimental setup, it is evident that bulk PZT (lead zirconate titanate) is bonded to an SOI (silicon-on-insulator) structure, resulting in the formation of a unimorph cantilever. Implementing SOI results in a flatter overall contour for the structure. Following the bonding of the bulk piezo-ceramic to silicon wafers, it can be produced using micromachining procedures. It was found that this piezoelectric device's utilization of silicon wafer bonding machinery was particularly compatible with MEMS technology. The performance of micropiezoelectricity harvesters has improved significantly over the past 10 years because of the use of KNN/Mn/KCT material in a transformer application. With increased dielectric, piezoelectric, and ferroelectric characteristics and a high Curie temperature, KNN is a potential lead-free piezoelectric material. Through the co-doping of KNN with Mn and $K_{5.4}$ $Cu_{1.3}$ Ta_{10} (KCT) sintering aids, this technique improves the densification of KNN. The KNN/Mn/KCT material greatly increases the piezoelectric constant and electromechanical coupling factor. This discovery suggests that modest mechanical-to-electrical energy sensors can be used in applications on a large scale.

Because of these properties, piezoelectric materials are fundamentally sensors and actuators. As a result, incorporating piezoelectric materials into MEMS allows unique functionality, particularly in basic microstructures [13].

1.8 KEY COMPONENTS OF INDUSTRY 4.0 IN MEMS

Advanced Manufacturing Techniques

Industry 4.0 technologies enable the use of modernized manufacturing techniques, such as additive manufacturing (3D printing) and micro/nano fabrication methods, to produce complex MEMS devices with high precision and efficiency. These techniques allow for rapid prototyping, customization, and the integration of multiple functionalities into a single device.

IoT-Enabled MEMS Sensors

MEMS sensors are critical components of the IoT ecosystem because they provide real-time data on a variety of characteristics. MEMS sensors may be combined with IoT platforms in Industry 4.0, allowing remote monitoring and control of physical systems. This integration enables MEMS-based devices and systems to have predictive maintenance, optimal performance, and increased efficiency.

Data Analytics (DA) and Machine Learning (ML)

MEMS devices provide a large quantity of data, which Industry 4.0 uses to derive useful insights. Manufacturers may spot patterns, anomalies, and trends in sensor data generated by MEMS devices, enabling proactive decision-making, quality control, and process optimization.

Smart MEMS Devices

The 4th Industrial Revolution encourages the development of smart MEMS devices capable of communicating with other devices, systems, and humans. These smart MEMS devices may be combined with wireless communication modules to provide real-time data sharing and control.

Quality Control and Testing

Industry 4.0 techniques, such as computer vision and automated inspection systems, can be applied to MEMS manufacturing processes to ensure quality control and efficient testing. These techniques help detect defects, measure critical dimensions, and validate the performance of MEMS devices, reducing human error and enhancing product reliability.

1.9 TESTING

MEMS are examined at several stages during the manufacturing and fabrication processes. These tests are critical for analyzing and verifying device performance, as well as parametrically and functionally assessing system output data. During the fabrication process, the broad majority of MEMS devices can be created using three fabrication processes: surface micromachining, bulk micromachining, and the molding process. MEMS devices are tested with automatic test equipment (ATE) [14] during wafer-level manufacture by monitoring both AC and DC characteristics on the wafer level [15,16]. This test procedure separates the wafer for both positive and negative dying by utilizing designs for checking circuits in chips, self-assess mechanisms, and scan chains that are similar to integrated circuits.

Following dicing and wire bonding to test power efficiency, excellent devices are finally packed. Ultimately, these packaged devices undergo parametric retesting to verify their overall functionality. Prior to their actual utilization, functional testing and configuration are imperative for all MEMS devices. During the calibration process, the inherent parameters or values associated with device performance are established. This is accomplished by subjecting the devices to predetermined pneumatic and electrical stimulation to compare their resulting output responses. If there is a difference between the measured and estimated values, the device is regarded as unacceptable; otherwise, it is approved as an acceptable device. Figure 1.4 depicts the detailed fabrication and testing stages for the mass manufacture of a MEMS pressure sensor. In this diagram, steps 6 and 12 are for device assembly, while steps 3, 7, 9, and 16 are for device packaging, steps 2, 8, 11, 14, and 15 are for testing, and steps 4, 5, 10, and 13 are for microlevel fabrication [17].

Testing is important for ensuring the performance and dependability of the device during the MEMS production process, but it is quite expensive [18]. According to MEPTEC (Microelectronics Packaging and Test Engineering Council), the testing cost of MEMS is 20–45% of the total cost of element fabrication. According to Masi and Cortese [19], the total testing process consumes 25–35% of the material cost.

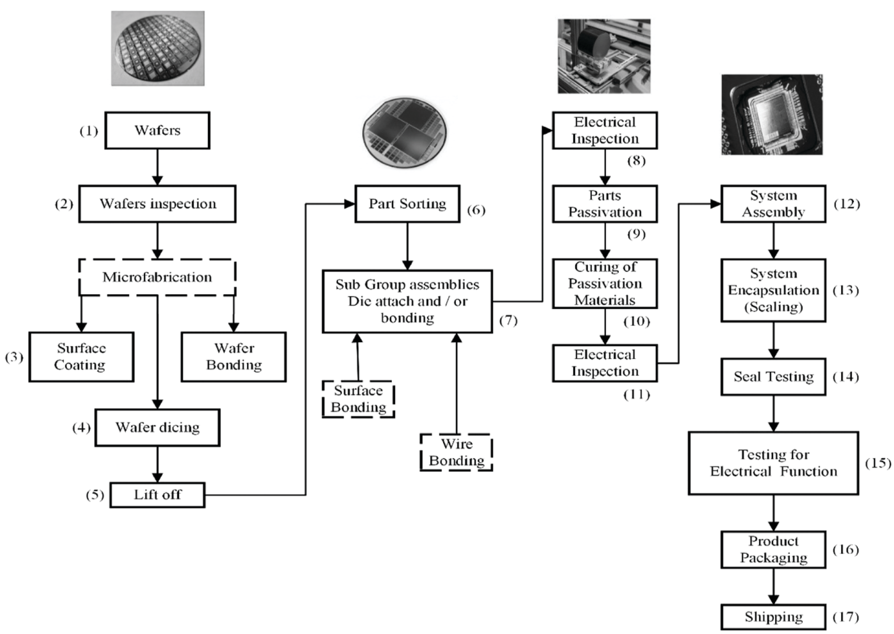

FIGURE 1.4 Detailed fabrication and testing stages for the mass manufacture of a MEMS pressure sensor [17].

According to Texas Instruments, for a digital micromirror device, the expense incurred for the final test, involving testing of the packaged device, amounts to 14% of the total cost. In contrast, the cost of inspecting the wafer at the wafer level accounts for 8% of the total cost [20]. However, according to the MIG (MEMS Industry Group) [21] for a MEMS test norm examination, testing costs at the stage of production can increase by up to 50% due to device design and sophistication.

1.9.1 The Differences among ICs versus MEMS Testing

Due to their multiphysics behavior, MEMS present numerous testing issues [22]. MEMS mechanical components are nonlinear, similar to analog electronic systems. MEMS requires analog and mixed signal tests [23], as well as verification and calibration. MEMS testing is complicated by their various qualities. A MEMS-based device's output, for example, is electrical in nature; yet, these signals are being generated by mechanical actuations in separate domains [24]. Consequently, utilizing electrical signals for testing purposes has proven advantageous in terms of controlling and observing mechanical components. In comparison to integrated circuits, MEMS technology is not as advanced. The monitoring and modeling of faults in MEMS devices are still in their early stages, lagging behind MEMS production and design. Despite being available since 1970, testing them remains a significant challenge. Conducting early-stage testing aids in production, fabrication, and even in packaging operations. The availability of analysis mechanism capable of identifying the causes of MEMS failures is crucial for upgrades and design improvements.

1.9.1.1 Technology Focus

IC Testing and Manufacturing

Integrated circuits involve the fabrication of electronic parts, such as transistors, resistors, and capacitors, on a semiconductor substrate. The focus is on creating functional electronic devices that perform logic, memory, or analog functions.

MEMS Testing and Manufacturing

MEMS technology involves the integration of mechanical and electromechanical components, such as sensors, actuators, and microstructures, on a substrate. The focus is on creating miniature mechanical systems that sense, measure, and actuate in response to physical stimuli.

1.9.1.2 Fabrication Processes

IC Manufacturing

ICs are primarily manufactured using photolithography, deposition, and etching processes to create the intricate electronic components on a semiconductor wafer (e.g., silicon).

MEMS Manufacturing

To manufacture mechanical and microelectronic components, MEMS devices are created utilizing various microfabrication techniques such as bulk micromachining, surface micromachining, and LIGA (lithography, electroplating, and molding).

1.9.1.3 Material Selection

ICs and MEMS

ICs are generally constructed of semiconductor materials like silicon that have been doped with certain impurities to provide the necessary electronic characteristics. To accomplish the requisite mechanical and electrical qualities, MEMS devices employ a variety of materials, including silicon, glass, metals, and polymers.

1.9.1.4 Testing Complexity

IC and MEMS Testing

IC testing focuses on electrical performance and functional testing to guarantee that digital logic or analog circuitry operates properly. Mechanical and electrical characterization are used in MEMS testing to confirm the mechanical behavior, sensing accuracy, and actuation capabilities of MEMS devices.

1.9.1.5 Device Architecture

ICs and MEMS

ICs are two-dimensional planar devices that are suited for electrical circuits and chip-scale integration. MEMS devices can have three-dimensional architectures because they frequently include movable or flexible components that enable mechanical functioning and sensing.

IC Packaging

ICs are typically encapsulated in hermetically sealed packages to protect them from environmental factors and ensure proper electrical connections.

MEMS Packaging

MEMS devices require specialized packaging to protect the delicate mechanical components and maintain their performance.

1.9.1.6 Application Areas

ICs

ICs are widely used in electronics, communication, computing, and other digital or analog applications.

MEMS

MEMS devices find applications in sensors, actuators, biomedical devices, automotive systems, consumer electronics, and IoT devices.

Failure modes [25] are as distinct as MEMS devices. Significant efforts have been expended in IC manufacturing to precisely assess and evaluate device performance in accordance with specified requirements. The fundamental distinction between testing MEMS and ICs is the environment. Integrated circuits undergo testing in diverse environments, subject to different temperature and moisture conditions in numerous scenarios. During the production process, somewhat similar testing and handling methods [26] are undertaken; nevertheless, the fundamental distinction arises from the unique environmental conditions in which the devices operate. In the

TABLE 1.1
Assessment Methodology of Microelectromechanical Systems [17]

Assessment Methodology	Digital Circuits	Analog Circuits	Microelectromechanical Systems
Fault method	Assertion, line, bus, redundancy, path, gate, behavioral, branch, cross-point, bridging, stuck-open, all are examples of delay	Catastrophic, hierarchical, macromodel, transistor, physical, behavioral, and parametric defects	Behavioral, electrothermal, and electromagnetic shorts and openings, structural fault level, parametric, operating functions, fatigue, and reliability models
Experiment method	BIST (built in self-test), HSPICE, VHDL, probabilistic, LFSR (linear feedback shift register), fault dictionary, are all examples of simulation techniques	ATPG, artificial neural network (ANN), HSPICE, SABRE, BIST (built-in self-test), Pole-Zero Analysis, VHDL-AMS	Device-level (FEM) and HDLs, as well as microelectronics system architecture, BIST, VHDL-AMS

test environment, any change can significantly impact the device's sensitivity and operation. The additional difficulty of mechanical motion necessitates extra caution during testing and handling (Table 1.1).

1.9.2 An Extensive Investigation into the Breakdown of MEMS Devicess

MEMS devices, unlike ICs, have dynamic characteristics; thus, unique methodologies and equipment are necessary for measuring mechanical responses at both the micro- and nanoscale. These technologies were used to analyze the device's behavior in depth, which aided in providing input to enhance the design. MEMS devices find extensive application across diverse global industries. Previously, there was an assumption that MEMS devices were part of the ICs family and might encounter similar failure concerns as ICs. On the other hand, MEMS devices have various breakdown mechanisms due to their complicated mechanical design and specific material, and they use explicit biasing approaches. Due to the complexity of these systems, the failure modes are classified into four groups.

Stationary MEMS devices that fall under Group A encompass stationary devices with no moving parts, including chemical sensors, microphones, and DNA sequencers. Particle contamination is the most common cause of failure in this group. Contaminated particles, being tiny and fundamentally nonelectrical, have a negative impact on device performance. Because particle contaminants are insulators, they cannot create an electrical connection across the device structure; due to this, contaminants are challenging to identify as short circuits.

Group B comprises of MEMS devices characterized by moving structures but lacking rubbing surfaces. This category includes comb drive accelerometers, where parts like hinges and microcantilever yoke regions are susceptible to fatigue failure. The focus of this group lies in examining structural failures caused by fatigue. Key components within this category include electrostatically actuated comb fingers featuring a perforated proof weight and a microcantilever featuring a notch. The critical aspect of examination involves the structural integrity of these components, specifically addressing the challenges posed by fatigue. In the case of electrostatically actuated comb fingers, the presence of a perforated proof weight adds complexity to the stress distribution. Similarly, the microcantilever, with its distinctive notch, becomes a focal point for analysis. The elevated stress levels at the notch are identified as the primary culprits triggering surface cracks on the microcantilever. These cracks, in turn, contribute to a restricted lifespan of the device and eventual failure attributable to fracture. Understanding and mitigating such fatigue-induced structural vulnerabilities are paramount for enhancing the reliability and longevity of MEMS devices in Group B.

MEMS devices having movable components and impact surfaces, including relays, valves, and thermal actuators, fall inside Group C. Such devices may be impacted by surface debris, fracture components, cracks [27], and so on. The device fails due to fracture caused by impact forces on the opposing structure.

Group D comprises of MEMS devices that involve translational components, rubbing motions, and impacting surfaces, such as micromirrors, optical shutters, and geared devices. The interaction of uneven surfaces in these devices leads to friction, causing material wear or the presence of debris. This friction-related wear and debris can result in various types of failures, including: (1) stiction of rubbing or contacting areas, (2) wear due to particles which impact motion tolerance, and (3) particle contamination (Table 1.2).

1.9.3 MEMS Device Reliability Testing

MEMS device functionality is determined by the structure's material, design, and actuating elements. As a result, the technology sector uses MEMS-based product reliability testing. MEMS device reliability testing is a significant challenge in bulk production and manufacturing. In this test, the failure analysis of the device under test (DUT) is conducted under particular circumstances that align with the device's application and within a particular duration. The product and application area greatly affect the causes of MEMS reliability failures. During the reliability test, it is necessary to employ an economical method for conducting prominent failure analysis to understand the characteristics of the DUT.

It is obvious that every type of MEMS device has failure processes caused by stiction, particle contamination, and other factors. As a result, understanding the causes of failure is far more significant than finding solutions to individual failures. Due to their distinct categories and nature, MEMS sensors and actuators exhibit separate

TABLE 1.2
Brief Details of Breakdowns, Causes, and Acquisition of Several MEMS Devices

Group	Application	Failure Cause	Reliability Testing	References
A	Nozzles, chemical sensors, microfluidics	Breakdown of dielectric	Flagellar microfluidic motors were tested for reliability under environmental settings. The rotational behavior was examined using a probabilistic mathematical model. The degradation time of the bacterial motor was anticipated to be 55 hours	[28]
A or B	Accelerometers	Structural changes and charging such as fracture, fatigue, change in friction	(a) The life duration was investigated practically for 1000 hours at temperatures ranging from 145°C to 200°C (b) Tytron 250 was utilized for fatigue testing. When the loads were increased from 2.05 to 2.83 GPa, the fatigue life ranged between 77.8k–14.8M cycles (c) Sandia National Laboratories created the SHiMMeR, which can test up to 256 MEMS parts at same time (d) X-ray diffraction was used to observe fatigue and creep. Stiction was measured using MIL-STD-883F. Wear was measured using DLC coating (Diamond Like Carbon), and contamination was measured using SAM (scanning acoustic microscope)	[29–31]
B	Pressure sensors	Structural changes like fracture, friction changes vibrations, shock, fatigue	It was discovered that fatigue can occur at equivalent stress and fracture levels in the constructed sensors functioning below the stress level (a) A three-axis gyroscope's reliability was evaluated under various shock loading circumstances (b) Temperature declines and changes in S/N ratio have also been recorded	[32–33]

TABLE 1.2 (Continued)
Brief Details of Breakdowns, Causes, and Acquisition of Several MEMS Devices

Group	Application	Failure Cause	Reliability Testing	References
	Gyroscopes	Vibrations and electric charge		
C	Thermal actuators	Vibration, shock, and structural deformation	(a) Testing for fracture on the beam using variable loads was analyzed using Weibull statistics (b) An experimental technique for observing malfunction due to electrical charge and creep was proposed for microswitches. (c) The effects of drawing voltages on beam bending and torsional modes were investigated. To determine the life duration, structural deformation and fatigue, experiments were done	[34–38]
	Microrelays	Structural deformation like fracture, electric charges, fatigue, shock, vibrations		
D	Electrostatic actuators	Structural deformation like fracture, electric charges, fatigue, shock, vibrations	The stochastic technique was used to analyze voltages in order to predict device life time	[39–42]
	Convex devices	Optical degradation, structural deformation like fracture, electric charges, fatigue, shock, vibrations	Texas Instruments (TI) created an optical analysis tool for DMD devices that inspects each pixel in the DMD array	
	Mechanical devices	Structural deformation like fracture, electric charges, fatigue, shock, vibrations	Simulations of sliding surfaces were used to avoid friction and wear	

TABLE 1.3
MEMS Mechanical Breakdown [17]

Mechanical Structural Breakdown					
Breakdown	Fatigue	Overload	Corrosion	Shock	
Reason	Structural damage due to periodic loading	Excessive stress	Oxidation, chemical reaction	Mechanical structure disorder, huge loading and drops	
Stiction					
Breakdown	Capillary forces	van der Waals forces	Residual stress	Chemical bond	Electrostatic charge
Reason	Capillary force is induced by a monolayer of moist or lubricant on the surfaces	Reactions of molecules at the surfaces of contact	Bending and deformation of structure during the release process	Surfaces contact bond	Electric potential at closed surfaces
Wear					
Breakdown	Surface fatigue	Corrosion	Lesion	Adherence	
Reason	Periodic loading rather than flat surfaces	Chemical synergy between two surfaces	Substance loss as a result of different surfaces sliding	interaction held within two surfaces	
Creep and Fatigue					
Breakdown	Thermal stress	Applied stress	Intrinsic stress		
Reason	Excess heating	Applied stress	Residual stress		

failure mechanisms. While in ICs, testing focuses solely on electrical parameters to identify defects, MEMS devices require comprehensive testing of both electrical and mechanical characteristics to address potential failures. Table 1.3 summarizes the MEMS failure causes throughout the reliability test.

1.9.4 MEMS Device Breakdown

These devices possess conductive or insulated flat layers, gears, hinges, cavities, and an electronic monitoring circuit. In contrast, multiple devices are integrated within a single chip. Moving on a single axis, MEMS architectures have been optimized to enable multiaxis movements. MEMS are becoming increasingly versatile in their design to accommodate a broader range of applications. Failure analysis plays a crucial role in comprehending the underlying causes of faulty device behavior throughout the testing operation, spanning from the wafer level to the last phase, i.e., packing of

TABLE 1.4
MEMS Electrical Breakdown [17]

Electric Circuit (Short and Open)				
Breakdown	Oxidation	Electromigration	ESD, high electric field	Degradation of dielectric material
Reason	Environment	Load variation	Overload	Capacitive discharges
Contamination				
Breakdown	Environment	Intrinsic (crystal growth)	Manufacturing-induced	
Reason	Variable temperature and humidity	Environment	Industry rough handling	

the device. The failure mechanism results in poor device performance due to defective interconnects and limited compatibility.

Table 1.3 lists the most common mechanical faults of MEMS devices. While MEMS failures extend beyond the mechanical aspects of the device, it is important not to overlook potential electrical failures. Contact failure, electro-migration, and dielectric degradation are the causes of these.

Table 1.4 lists the most common electrical breakdowns of MEMS devices. MEMS devices are the most difficult to test due to their multiphysics behavior. Recently, these gadgets have been tested using functional specifications. Variations can influence the specifications of the device in the manufacturing process, which is a detrimental condition in MEMS testing because process fluctuations might introduce a lot of faults. In addition, the device's specifications are diverse and the device's measurements at different testing stages can prolong the overall test duration. Consequently, a wide range of test methodologies is necessary to measure the attributes of performance of different devices, compelling manufacturers to acquire diverse test equipment for each product. To validate the DUT specifications, fast transient test approaches are necessary. Specifications in multistructured complicated devices are frequently altered due to fabrication process drifts and faults. Specification violations have occurred due to normal manufacturing changes for devices designed near the specification limits. Process monitoring systems have detected process drifts at an early point in the production phase. Defects serve as the root causes of catastrophic failures, which can be identified through parametric and functional failure analysis tests.

Some fields like shape memory alloys (SMAs) and shape memory polymers (SMPs) for micro- and nanotechnology applications have the capabilities to remember the predeformed shape. Current science has various techniques to form micro-sized as well as nano-sized SMA and SMP materials, usually in the form of thin films. The growing future of 3D printing technology and devices like micropumps and microactuators shows enormous potential in shape memory material nanotechnology [43]. Table 1.5 summarizes the issues in the SMAs-Based Thin Films and SMPs.

TABLE 1.5
Challenges in the SMAs-Based Thin Films and SMPs

S. no.	SMAs-Based Thin Films	Challenges in SMPs
1.	Problems with fatigue	Interaction between heat production and thermo-response, as well as structural and polymer network size issues in SMPs for nanotechnology applications
2.	Temperature-driven changes demonstrate significant hysteresis, which is affected by SMA preparation procedures and material composition	Difficulties with tensile testing (SMA/SMP). Internal stress created after the sampling process might lead free-standing samples to bend. These bent samples not only affect the test outcome, but they are also readily destroyed when clamped in the tensile test device
3.	Low actuation speed	Experiments on the shape memory effect and superelasticity in SMA/SMP at the nanoscale remain extremely difficult. For example, during the phase transformation, it is difficult to discern grain evolution/changes inside individual grains
4.	Challenges in separating the martensite and R-phase temperature domains	
5.	Handling the R-phase transformation temperatures and the minimal transformation strain poses challenges. Such challenges are limiting the R-phase's extensive adoption in current nanotechnology applications	
6.	The surface preparation required to avoid the release of poisonous Ni using the arc deposition process is a difficult undertaking because this technique could lead to the generation of microdroplets and also it requires precise control of the properties of plasma and the vacuum arc	
7.	Precise control of composition of ultrathin films during its preparation, especially at the nanoscale level, is a major issue because at nanoscale dimensions, the transformation behavior can significantly deviate	

TABLE 1.5 (Continued)
Challenges in the SMAs-Based Thin Films and SMPs

S. no.	SMAs-Based Thin Films	Challenges in SMPs
8.	Determination of nanoscale sample phase transformation temperatures. This may need the use of advanced devices such as nanocalorimeters. As a result, characterizing the shape memory and superelastic responses of micro-/nano-sized SMA and SMP samples is a difficult challenge. It necessitates precise measurements of forces and changes in displacement	
9.	Difficulties with tensile testing (SMA/SMP). Internal stress created after the sampling process might lead free-standing samples to bend. These bent samples not only affect the test outcome, but they are also readily destroyed when clamped in the tensile test device	
10.	Experiments on the shape memory effect and superelasticity in SMA/SMP at the nanoscale remain extremely difficult. For example, during the phase transformation, it is difficult to discern grain evolution/changes inside individual grains	

1.10 ISSUES DURING TESTING OF MEMS

MEMS are miniature devices that integrate mechanical and electrical components on a single chip. Testing MEMS devices can present several challenges due to their small size, complex structures, and the combination of mechanical and electrical functionality. Issues that can arise during the testing of MEMS are described below.

1.10.1 Fabrication Failures

Microfabrication techniques are essential in the manufacturing of MEMS. These techniques enable the precise and controlled fabrication of microstructures, sensors, actuators, and integrated circuits on a small scale. MEMS devices often involve complex and intricate structures that require specialized microfabrication processes as shown in Figure 1.5. Some common microfabrication techniques used in MEMS manufacturing described next.

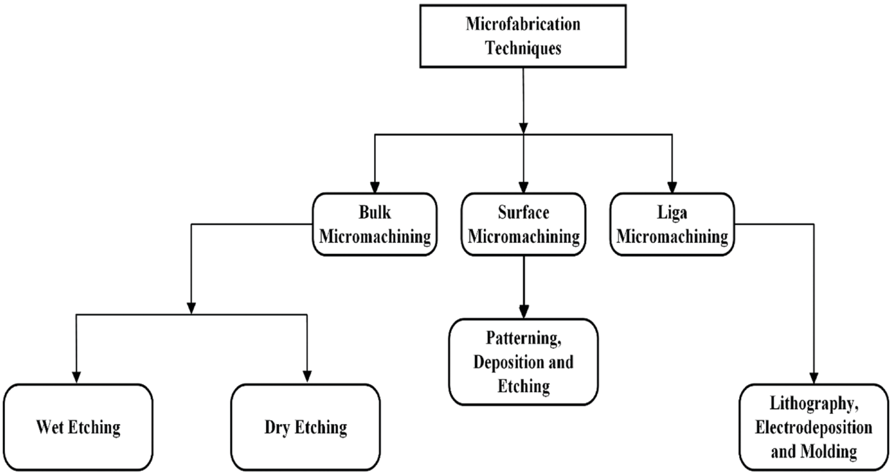

FIGURE 1.5 Microfabrication techniques for MEMS.

1.10.2 Photolithography

Photolithography is a crucial microfabrication process that uses light exposure and photoresist materials to form patterns on a substrate. It entails putting a photosensitive substance (photoresist) on the substrate, exposing it to light selectively through a photomask, and developing it to make the pattern of choice. The key features and layers of MEMS devices are created via photolithography.

1.10.3 Etching

Etching is the technique of removing material from a substrate selectively to form shapes and features. Wet etching involves soaking the substrate in a chemical solution that dissolves certain compounds, whereas dry etching removes material using plasma or ion beams. Etching is a technique used to determine the forms of MEMS components such sensors, actuators, and microfluidic channels.

1.10.4 Deposition

This requires depositing thin layers of materials onto a substrate in order to construct the desired MEMS structure. In MEMS production, typical deposition technologies include chemical vapor deposition (CVD), physical vapor deposition (PVD), and electroplating.

1.10.5 Oxidation

The process of oxidation is used to develop thin layers of oxide on a substrate, commonly silicon. This approach is utilized to build insulating layers as well as sacrificial layers for the release of MEMS systems.

1.10.6 Lithography

Lithography, like photolithography, involves producing patterns on a substrate via masks and light exposure. Lithography, on the other hand, is often utilized for patterning at higher sizes than photolithography, making it appropriate for some MEMS applications.

1.10.7 Bonding

Bonding methods are employed in MEMS devices to join substrates or layers of various materials. Bonding methods often utilized in MEMS production include wafer bonding, fusion bonding, and anodic bonding.

1.10.8 Surface Micromachining

Surface micromachining is the process of creating MEMS structures on a substrate's surface. Thin layers of material are deposited and patterned, and then further layers are created on top to make the required structures.

1.10.9 Bulk Micromachining

To construct 3D MEMS structures, bulk micromachining entails etching into the bulk of the substrate. This method is used to create gadgets that have moving components and mechanical functioning. These microfabrication techniques, along with other advanced processes, enable the precise and repeatable fabrication of MEMS devices with complex functionalities, making MEMS technology applicable in a wide range of industries, including electronics, healthcare, automotive, and consumer electronics.

Surface micromachining [44,45] is a direct evolution of the technology employed in manufacturing VLSI (very large-scale integration) devices using CMOS (complementary metal-oxide-semiconductor) technology. In such systems, thin layers of conductor, insulator, semiconductor, and passivation materials are deposited on doped silicon wafer substrates. In VLSI systems, these layers are coated, etched, and patterned to create highly cohesive electronic devices with smaller component sizes.

On the other hand, in surface micromachined MEMS, these layers are etched and shaped to construct electromechanical components that act as sacrificial layers, enabling the passage of mechanical layers. By utilizing CMOS-compliant processes and materials, the coupling of mechanical with electronic devices required for data processing and power transmission can be achieved, adding value to the overall system. Examples of commercially available extensively integrated micromachined surface devices include microaccelerometers chips used to control the activation of vehicle airbags and mirror arrays utilized in movable projectors. Figure 1.6 shows a surface micromachines process, which depicts amorphous, silicon carbide, and photoresist layers. These micromachining processes are often confined to thickness layers below 5 mm, limiting the ability to build devices that can distribute significant mechanical force.

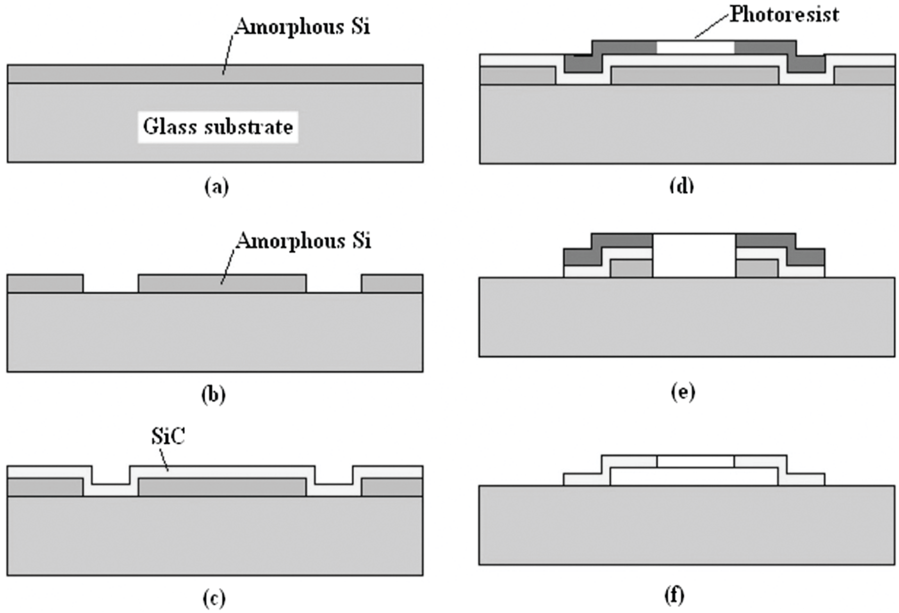

FIGURE 1.6 Typical surface micromachines process [47].

Bulk micromachining [46] involves the direct etching of a silicon wafer or another substrate. When integrated electrical functionalities are desired, the microelectronic components on the topmost layer of the silicon wafer are typically fabricated using CMOS methods. The remaining sides of the wafer are then used for bulk micromachining to construct mechanical elements like slim diaphragms, beams, or fluid flow tunnels. For many years, this technology has been employed to produce compact optical pressure sensors. Capacitive or piezoresistive techniques are utilized to gauge the bending of a slender membrane on an entire micromachined cavity. The cavity is sealed and vented to form a relative or absolute pressure sensor. Figure 1.7 provides a schematic representation of bulk micromachining that allows for the use of wider and thicker features compared to surface micromachining, primarily due to the silicon substrate serving as the foundation for physical system components. This is particularly advantageous in MEMS applications that require higher mechanical forces or strength levels, or in fluid applications like inkjet printer nozzles, where flow through small channels is achieved by surface micromachining and can lead to significant losses.

1.11 MOLDING PROCESS ISSUES

The production of the mechanical components of MEMS devices using a micromold is the third most prevalent industrial method. The term "LIGA" is derived from the German abbreviations for the primary process phases: lithography, galvanoforming (electroforming), and molding. The fundamental procedure entails lithography to

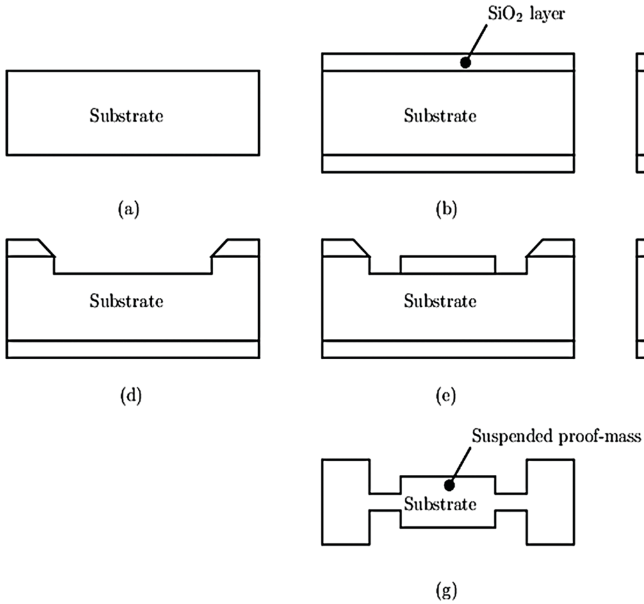

FIGURE 1.7 Schematic diagram of bulk micromachining [48].

create polymer molds [47] and subsequently electroplating metals into the cavities of these molds. The concept of using a molding mechanism is not limited to the deposition method. Certain substances, includes polycrystalline silicons and silicon carbides, can be processed using CVD and refractory ceramic structures generated by the sludge process approach.

1.11.1 Dicing Issues

The swift advancement and increased accessibility of cost-effective wafers, novel electrode materials, and thin-film targets with superior electrical output and compatibility with semiconductor processing pose challenges for MEMS devices. Achieving elevated temperature stability and resistance to perpetual deformation in MEMS devices is crucial. However, these requirements exceed the capabilities of delicate mechanisms. The MEMS industry is currently dealing with the most serious issue: device breakage during dicing.

1.11.2 Designing Issues

The ease and certainty with which complicated devices can be developed without the need for extensive prototypes is an essential goal for the continuous advancement of technology as well as the economic development of the microelectronic sectors [48]. The reliability of the existing modeling tools, along with the extensively characterized electrical properties of the materials used and the manufacturing techniques employed, greatly facilitate the design process of microelectronic devices. To unlock the potential of low component price and large-scale manufacturing in MEMS, it is crucial to adopt a similar design approach. In response to this demand, numerous simulation tools have been developed and are commercially available, catering specifically to the design of MEMS devices. The abundance of common testing techniques and material characteristics libraries has hampered the usability of simulation and design methodologies. Despite the recognition of the need for such capabilities as early as 1986, convincing progress has been only recently made. The issues lie in the fact that the properties of microfabricated materials are highly affected by the fabrication process and the scale of the structures involved. Mechanical properties estimated at the micro scale can significantly differ from bulk samples of material measured at the macro size. Moreover, non-scale-dependent properties like elastic modulus and density [49] can be altered due to vapor deposition gas, substrate effects, and nonequilibrium microstructure. To fully harness the advantages of rapid and systematic simulators for MEMS advancement, methodologies are being developed to establish connections between the material characteristics achieved and the assemblies and materials used. Figure 1.8 provide a schematic diagram of MEMS design. The initial

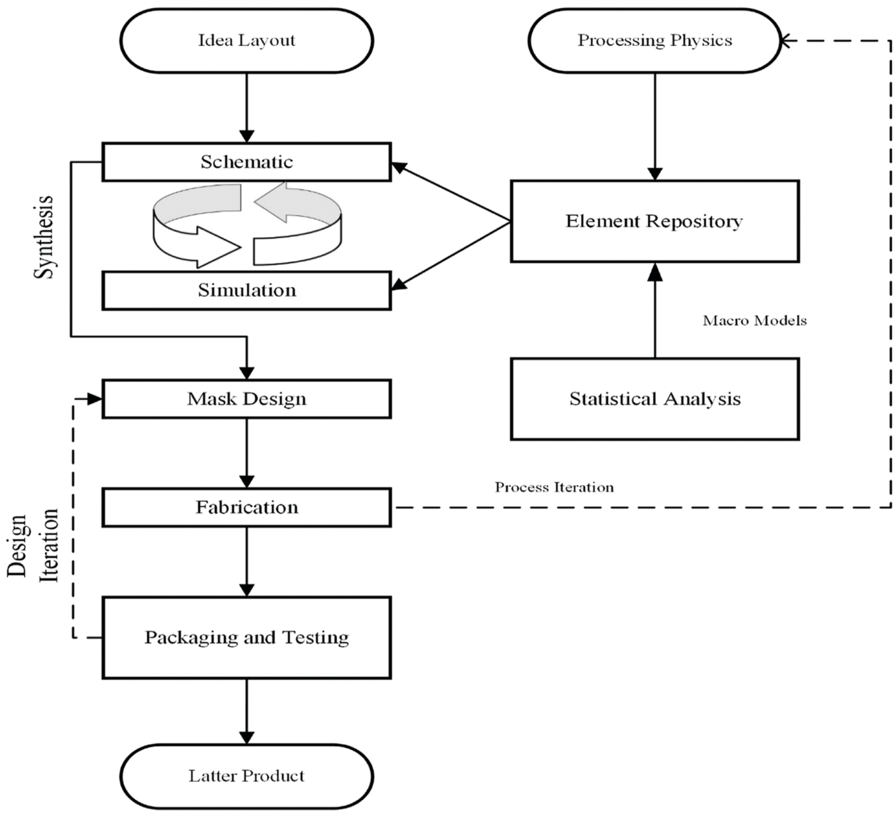

FIGURE 1.8 Schematic diagram of MEMS design [3].

step involves establishing graded testing protocols to characterize the structural properties of micromechanical manufactured materials using procedures and criteria specific to the intended applications. This will enable the creation of verified product and process databases, as well as the linkage of manufacturing paths and assets, to enable simulation-based design. Various studies have focused on key design issues such as elastic properties, strength characterization, bond strength, residual stress, fatigue, adhesion, surface forces, and tribology.

1.11.3 Testing at Wafer Level Issues

Comparing packaging of MEMS devices at the wafer level to bare die MEMS devices is not appropriate. Testing every sample die on the whole wafer substrate is challenging and time-consuming, which may result in inadequately testing wafer-level packaged products. Furthermore, there are additional obstacles to overcome, for example, consider the limited area between wire bonding and the handling challenges. The fundamental challenge in testing packed wafer-level devices is developing a system capable of evaluating numerous dye in a single test cycle.

1.11.4 Packaging and Handling

MEMS devices have a tendency to be sensitive and must be handled with care during testing and packing. They are susceptible to environmental conditions including humidity, tremors, and temperature. During testing, certain care must be taken to safeguard the equipment from damage or contamination.

1.11.5 Test Structure Design

Creating an appropriate test framework for MEMS devices may be difficult. Access to both electrical and mechanical interfaces must be provided by the test structure, permitting functional testing of the device's components. It might be difficult to design adequate test structures that do not interfere with the device's performance or modify its behavior.

1.11.6 Sensor Calibration

Calibration is required for many MEMS devices, such as gyroscopes, pressure sensors, and accelerometers, to guarantee precise measurement. Calibration is the process of analyzing the device's reaction and changing its output to meet established reference values. It can be difficult to develop calibration techniques and equipment that can manage the small size and complicated behavior of MEMS sensors.

1.11.7 Electrical and Mechanical Coupling

Mechanical and electrical components of MEMS devices frequently interact in complex ways. Testing these devices necessitates assessing both electrical and mechanical performance at the same time. During testing, it might be difficult to identify and isolate the impacts of each component.

1.11.8 Test Equipment Limitations

Because of their small size and particular properties, conventional test equipment may not be adequate for evaluating MEMS devices. To precisely access and test MEMS devices, specialized equipment such as microprobes, microscopes, and environmental chambers may be necessary. Such technology can be costly and may need specialized knowledge to operate efficiently.

1.11.9 Yield and Reliability

MEMS devices are mass-produced, therefore maintaining high yield and dependability is critical. Testing assists in identifying any manufacturing flaws or variances that may have an impact on the performance or dependability of the devices. Developing effective testing procedures to reduce false positives and false negatives and increase yield can be difficult.

1.12 CHALLENGES ENCOUNTERED DURING THE MANUFACTURING OF MEMS DEVICES

MEMS device manufacture presents various obstacles due to its complicated architecture, tiny size, and integration of multiple components. Figure 1.9 depicts the MEMS manufacturing procedures. Some frequent issues faced during MEMS device production include:

Process complexity: MEMS production entails complex techniques such as deposition, etching, lithography, and bonding. It might be difficult to coordinate these activities and ensure their compliance with the required device structure. To obtain the necessary mechanical and electrical properties of the device, the manufacturing procedures must be properly regulated.

Miniaturization and precision: MEMS devices are often made at the microscale, necessitating tremendous accuracy. It can be difficult to achieve the appropriate feature sizes, such as submicron diameters, and to maintain consistency across large batches of devices. The sensitivity to process changes is likewise increased by miniaturization, making process control crucial.

Material selection and compatibility: MEMS devices frequently necessitate the use of several materials with varied mechanical, electrical, and chemical capabilities. It might be difficult to select appropriate materials that can endure manufacturing procedures and perform consistently in the device's intended environment. Material compatibility, stress mismatch, and variances in thermal expansion must all be carefully examined.

Packaging and hermetic process: It is an enormous challenge to package MEMS devices so that they are protected from external conditions while remaining functional. Moisture, particulates, and impurities may all influence the performance

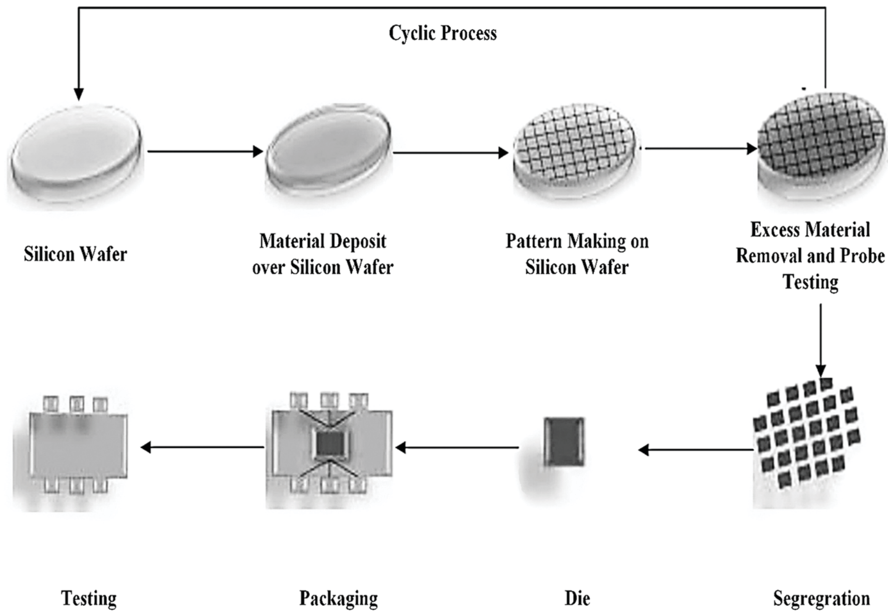

FIGURE 1.9 MEMS manufacturing process [3].

and dependability of MEMS devices. It is critical yet difficult to achieve a hermetic seal to prevent impurities from entering the device without affecting its mechanical or electrical behavior.

Yield improvement: MEMS production frequently includes batch processing, in which a large number of devices are produced on a wafer at the same time. Due to process differences, material difficulties, and complicated device designs, achieving high yield and eliminating defects over the whole wafer can be difficult. To improve production efficiency, yield enhancement methods such as defect identification, process optimization, and statistical process control are required.

Testing and characterization: Validating the functionality and performance of MEMS devices is essential during manufacturing. However, testing complex MEMS structures can be challenging due to their small size, the integration of mechanical and electrical components, and the need for specialized equipment. Developing reliable and efficient testing methodologies to ensure the quality and reliability of each device can be demanding.

Future challenges faced by the MEMS failure analysis in six distinct categories based on their applications are sensors, actuators, RF MEMS, optical MEMS, microfluidic MEMS, and bio-MEMS. A major challenge for failure analysis common to all these categories is to develop a technique that performs multiple measurements on single and several devices in parallel at the same time. For sensors, a major hurdle is simulating the chemical behavior of the devices designed to sense. Other major areas are packaging and depackaging/decapping. In actuators, the challenge is to determine the critical temperature, whereas in the case of RF MEMS and optical MEMS the challenges include analysis of surface roughness, nondestructive analysis of adhesion or stiction failures, and charge accumulation.

Similarly, challenges for microfluidic MEMS and bio-MEMS include functional and structural analysis, biocompatibility, etc. Therefore, nondestructive tools and techniques must be developed that can assess the failure mechanisms on a large scale. Therefore, the industries, academia, and various government institutions must work together to create the knowledge base that is required to develop such tools and techniques [50].

1.12.1 Biomedical Applications of MEMS Devices and Their Issues and Challenges

Technical advancements in the field of biomedical MEMS, known as bio-MEMS, have been growing over the last two decades as efficient microscale as well as nanoscale tools for the precise, fast, highly sensitive, and cost-efficient alternatives for disease diagnoses and processing. In Figure 1.10, bio-MEMS includes the biology discipline and MEMS, and it has been developed and designed for the upliftment of diagnostic devices and biosensors, surgical tools, microfluidic devices, inertial sensors for biomechanics, ultrasound transducers, drug-delivery systems, and lab-on-a-chip (LOC) devices [51].

Lab-on-a-chip (LOC) devices: MEMS-based lab-on-a-chip devices enable the miniaturization and integration of complex laboratory processes, such as sample preparation, mixing, separation, and detection, onto a single microfluidic chip. LOC devices have applications in medical diagnostics, DNA analysis, drug discovery, and point-of-care testing, offering faster, cost-effective, and portable solutions.

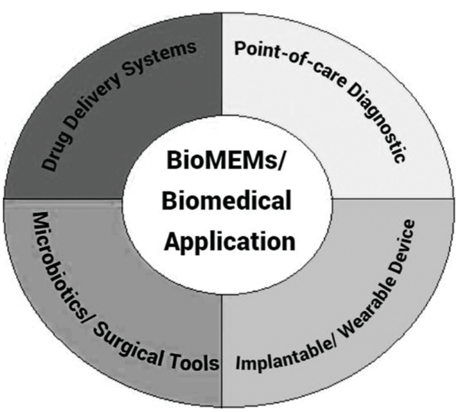

FIGURE 1.10 MEMS applications in the biomedical field.

Biosensors: MEMS biosensors are capable of detecting and quantifying biological molecules, cells, and pathogens with high sensitivity and specificity. Submicron-range MEMS are being developed so that they can be useful in miniaturized diagnostic devices. These biosensors find applications in detecting biomarkers for various diseases, monitoring glucose levels, analyzing DNA sequences, and identifying infectious agents. Specific MEMS components, such as micropumps, micromixers, and microvalves, are used in these applications to regulate and analyze liquids, fluids, and chemical processes. For disease diagnosis, body fluids such as blood, serum, urine, saliva, or liquid biopsies are frequently introduced into the microchip. Biosensors have been used to detect and monitor glucose, hemoglobin, urea, amino acids, bodily gases, viruses, bacteria, and cancer biomarkers. BioMEMS, on the other hand, can be employed as wearable or implanted devices for fixed and continuous measurement of blood oxygen, intraocular or aortic pressure, electrical impulses, or metabolic processes. Wearable MEMS have become an important component of chronic patients' everyday life since they may remotely track vital statistics such as blood and intracranial pressure, glucose levels, heart and respiration rate, body temperature, and O_2 saturation [52–60].

Implantable devices: MEMS-based implantable devices are used in various medical applications, including neural prosthetics, pacemakers, drug-delivery systems, and intraocular pressure sensors for glaucoma patients. Their miniaturization allows for minimally invasive procedures and reduced risk of complications.

Microfluidic devices: MEMS-based microfluidic devices enable precise manipulation and control of fluids at the microscale. They are used for applications such as cell sorting, drug delivery, organ-on-chip models, and analysis of single cells, contributing to advancements in personalized medicine and drug development.

Inertial sensors for biomechanics: Wearable gadgets use MEMS accelerometers and gyroscopes to track human movements and measure physical activity. These sensors are used to monitor sports and fitness, as well as for rehabilitation and gait analysis.

Ultrasound transducers: MEMS-based ultrasound transducers enable the creation of smaller, portable ultrasound devices while improving imaging capabilities. They have medical imaging uses, such as point-of-care ultrasonography and less invasive treatments.

Drug-delivery systems: MEMS-based drug-delivery systems have the ability to precisely regulate drug release rates and dosages, hence improving therapeutic efficacy and decreasing adverse effects. They have uses in cancer therapy, insulin administration for diabetic management, and other controlled-release medicines.

Bio-MEMS: Bio-MEMS have demonstrated tremendous potential in numerous aspects of infectious illness detection, diagnosis, monitoring, and treatment, including during the coronavirus (COVID-19) pandemic caused by the SARS-CoV-2 virus.

Diagnostic testing: Bio-MEMS-based lab-on-a-chip devices have been developed for rapid and sensitive detection of the SARS-CoV-2 virus. These devices use microfluidic technology to miniaturize and integrate the different steps of the diagnostic process, enabling faster and point-of-care testing. Polymerase chain reaction (PCR) amplification, isothermal amplification, and CRISPR-based detection methods have been integrated into bio-MEMS devices for COVID-19 testing.

Wearable health monitoring: Wearable bio-MEMS devices, such as biosensors and microfluidic patches, have been explored for continuous monitoring of biomarkers related to COVID-19, such as body temperature, respiratory rate, and cytokine levels. These devices can provide early indications of infection or disease progression and help in monitoring patient health remotely.

1.13 CONCLUSION AND FUTURE SCOPE

The most significant improvements in MEMS could come from technological advancements that allow for the production of larger instruments at unit prices comparable to those of current microelectronics. Mesoscale devices, which possess greater capacity and strength than modern MEMS, present an opportunity in Industry 4.0 to expand the range of materials and refine fabrication procedures. Material sciences must focus on improving methodologies for material selection and characterization in the context of microfabricated structures. While some testing tools and methods for IC technology can be applied for testing in MEMS, there is a need for new diagnostic tools to address specific challenges. The cost of testing must be reduced and value-added techniques implemented in Industry 4.0 and 5.0 to facilitate the adoption of MEMS as an emerging technology. Standard electrical testing is insufficient to capture MEMS devices' mechanical behavior, necessitating specialized techniques for flow, motion, and pressure testing. As device complexity increases with multi-function integration, testing becomes more challenging. Different testing approaches are employed throughout MEMS development's design, prototype, and production cycles. In Industry 4.0, Design for Test (DFT) is crucial to address cost and time-to-market pressures, and a comprehensive understanding of testing requirements is essential during the design phase to ensure cost-effectiveness and reliable prediction of device performance. Resolving these challenges in Industry 4.0 requires a holistic view of MEMS as a system, ensuring that material solutions align with the entire production process and application needs.

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